System Al–Zr

Solution Phases:

Liquid, bcc–$A2$, fcc–$A1$, hcp–$A3$

Compounds:

Al_2Zr, Al_2Zr_3, Al_3Zr, Al_3Zr_2, Al_3Zr_5, Al_4Zr_5, $AlZr$, $AlZr_2$, $AlZr_3$

Modelling:

Liquid	:	Substitutional, Redlich–Kister
bcc–$A2$	:	Substitutional, Redlich–Kister
fcc–$A1$	:	Substitutional, Redlich–Kister
hcp–$A3$	:	Substitutional, Redlich–Kister
Al_2Zr	:	Stoichiometric, $(Al)_2(Zr)$
Al_2Zr_3	:	Stoichiometric, $(Al)_2(Zr)_3$
Al_3Zr	:	Stoichiometric, $(Al)_3(Zr)$
Al_3Zr_2	:	Stoichiometric, $(Al)_3(Zr)_2$
Al_3Zr_5	:	Stoichiometric, $(Al)_3(Zr)_5$
Al_4Zr_5	:	Stoichiometric, $(Al)_4(Zr)_5$
$AlZr$	:	Stoichiometric, $(Al)(Zr)$
$AlZr_2$	:	Stoichiometric, $(Al)(Zr)_2$
$AlZr_3$	:	Stoichiometric, $(Al)(Zr)_3$

Assessor and Date:

N. Saunders (1991)

Thermodynamic properties of the solution and compound phases ($J.mol^{-1}$)

Phase Al_2Zr

$$G^{o}T - 2.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - H_{Zr}^{o,hcp-A3}(298.15\ K) = -137430.0 + 25.44\ T + 2.0\ GHSER_{Al} + GHSER_{Zr}$$

Phase Al_2Zr_3

$$G^{o}T - 2.0\ H_{Al}^{o,fcc-A1}(298.15\ K) - 3.0\ H_{Zr}^{o,hcp-A3}(298.15\ K) = -192135.0 + 33.0\ T + 2.0\ GHSER_{Al} + 3.0\ GHSER_{Zr}$$

Phase Al_3Zr

$$G^{\circ}T - 3.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -162500.0 + 28.92\ T + 3.0\ GHSER_{Al} + GHSER_{Zr}$$

Phase Al_3Zr_2

$$G^{\circ}T - 3.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 2.0\ H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -234700.0 + 44.1\ T + 3.0\ GHSER_{Al} + 2.0\ GHSER_{Zr}$$

Phase Al_3Zr_5

$$G^{\circ}T - 3.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 5.0\ H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -217488.0 + 48.72\ T + 3.0\ GHSER_{Al} + 5.0\ GHSER_{Zr}$$

Phase Al_4Zr_5

$$G^{\circ}T - 4.0\ H_{Al}^{\circ,fcc-A1}(298.15\ K) - 5.0\ H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -369000.0 + 62.55\ T + 4.0\ GHSER_{Al} + 5.0\ GHSER_{Zr}$$

Phase AlZr

$$G^{\circ}T - H_{Al}^{\circ,fcc-A1}(298.15\ K) - H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -89000.0 + 17.0384\ T + GHSER_{Al} + GHSER_{Zr}$$

Phase $AlZr_2$

$$G^{\circ}T - H_{Al}^{\circ,fcc-A1}(298.15\ K) - 2.0\ H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -100125.0 + 17.553\ T + GHSER_{Al} + 2.0\ GHSER_{Zr}$$

Phase $AlZr_3$

$$G^{\circ}T - H_{Al}^{\circ,fcc-A1}(298.15\ K) - 3.0\ H_{Zr}^{\circ,hcp-A3}(298.15\ K) = -108000.0 + 22.38\ T + GHSER_{Al} + 3.0\ GHSER_{Zr}$$

Phase bcc-$A2$

$$L_{Al,Zr:\square}^{0,bcc-A2} = -122300 + 32\ T$$

$$L_{Al,Zr:\square}^{1,bcc-A2} = -11000$$

$$L_{Al,Zr:\square}^{2,bcc-A2} = 15000$$

Phase fcc–A1

$$L^{0,\mathrm{fcc-A1}}_{\mathrm{Al,Zr:\square}} = -120000 + 30\ T$$

$$L^{1,\mathrm{fcc-A1}}_{\mathrm{Al,Zr:\square}} = -10000$$

$$L^{2,\mathrm{fcc-A1}}_{\mathrm{Al,Zr:\square}} = 15000$$

Phase hcp–A3

$$L^{0,\mathrm{hcp-A3}}_{\mathrm{Al,Zr:\square}} = -122300 + 32\ T$$

$$L^{1,\mathrm{hcp-A3}}_{\mathrm{Al,Zr:\square}} = -8000$$

$$L^{2,\mathrm{hcp-A3}}_{\mathrm{Al,Zr:\square}} = 17000$$

Phase liquid

$$L^{0,\mathrm{liquid}}_{\mathrm{Al,Zr}} = -125000 + 35\ T$$

$$L^{1,\mathrm{liquid}}_{\mathrm{Al,Zr}} = -10000 + 5.57\ T$$

$$L^{2,\mathrm{liquid}}_{\mathrm{Al,Zr}} = 15750$$

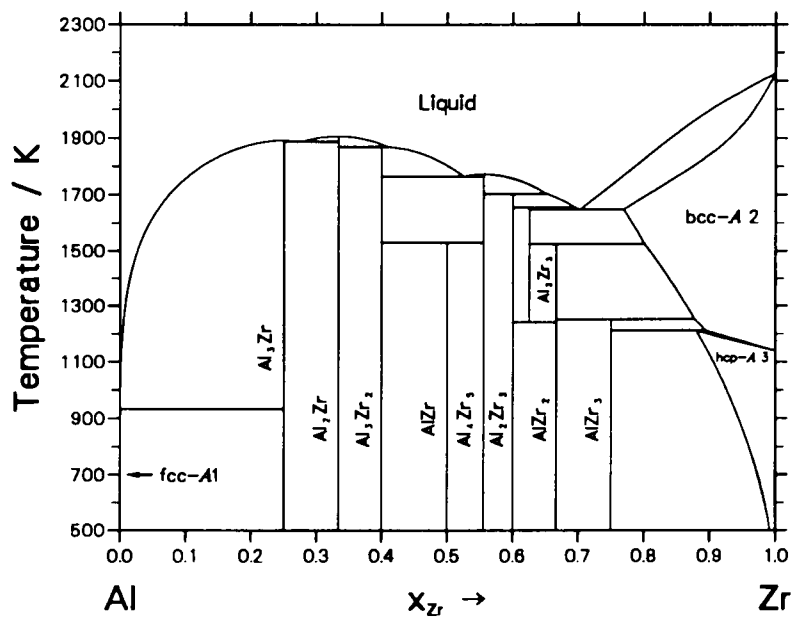

Table I - Invariant Reactions.

Reaction	Type	Compositions x_{Zr}			T / K
Liquid ⇌ fcc-$A1$ + Al_3Zr	Peritectic	.0002	.0008	.250	933.8
Al_3Zr ⇌ Liquid	Congruent	.250	.250	...	1893.0
Liquid ⇌ Al_3Zr + Al_2Zr	Eutectic	.277	.250	.333	1888.0
Al_2Zr ⇌ Liquid	Congruent	.333	.333	...	1907.0
Liquid + Al_2Zr ⇌ Al_3Zr_2	Peritectic	.410	.333	.400	1869.1
Liquid ⇌ Al_3Zr_2 + Al_4Zr_5	Eutectic	.525	.400	.556	1765.3
Al_4Zr_5 ⇌ Liquid	Congruent	.555	.555	...	1773.0
Liquid + Al_4Zr_5 ⇌ Al_2Zr_3	Peritectic	.653	.556	.600	1701.8
Liquid + Al_2Zr_3 ⇌ Al_3Zr_5	Peritectic	.696	.600	.625	1655.8
Liquid ⇌ Al_3Zr_5 + bcc-$A2$	Eutectic	.702	.625	.768	1648.7
Al_2Zr_3 + bcc-$A2$ ⇌ $AlZr_2$	Peritectoid	.600	.801	.666	1524.2
Al_3Zr_5 ⇌ Al_2Zr_3 + $AlZr_2$	Eutectoid	.625	.600	.666	1241.7
$AlZr_2$ + bcc-$A2$ ⇌ $AlZr_3$	Peritectoid	.667	.877	.750	1252.3
$AlZr_3$ + bcc-$A2$ ⇌ hcp-$A3$	Peritectoid	.750	.894	.879	1212.2
Al_3Zr_2 + Al_4Zr_5 ⇌ AlZr	Peritectoid	.400	.556	.500	1529.5

Table II:a - Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group	Sub-lattices		Comments
Al-Zr	(Al)	Cu	$cF4$ $Fm\bar{3}m$	M	4	
	(Zr)	Mg	$hP2$ $P6_3/mmc$	M	2	
	(Zr)	W	$cI2$ $Im\bar{3}m$	M	4	
	AlZr	BCr	$oC8$ $Cmcm$	Al Zr	4 4	
	$AlZr_2$	$InNi_2$	$hP6$ $P6_3/mmc$	Al Zr_1 Zr_2	2 2 4	

Table II:b - Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group	Sub-lattices		Comments
	$AlZr_3$	$AuCu_3$	*cP4* *Pm3̄m*	Al Zr	1 3	
	Al_2Zr	$MgZn_2$	*hP12* *P6₃/mmc*			
	Al_2Zr_3	Al_2Zr_3	*tP20* *P4₂/mnm*	Al Zr_1 Zr_2 Zr_3	8 4 4 4	
	Al_3Zr	Al_3r	*tI16* *I4/mmm*	Al_1 Al_2 Al_3 Zr	4 4 4 4	
	Al_3Zr_2	Al_3Zr_2	*oF40* *Fdd2*	Al_1 Al_2 Al_3	8 16 16	
	Al_3Zr_4	Al_3Zr_4	*hP7* *P6̄*	Al Zr_1 Zr_2 Zr_3	3 1 1 2	
	Al_3Zr_5	Mn_5Si_3	*hP16* *P6₃/mcm*	Al Zr_1 Zr_2	6 4 6	not in Masl
	Al_3Zr_5	Si_3W_5	*tI32* *I4/mcm*	Al_1 Al_1 Zr_1 Zr_2	4 8 4 16	
	Al_4Zr_5	Ga_4Ti_5	*hP18* *P6₃/mcm*	Al_1 Al_2 Zr_1 Zr_2	2 6 4 6	not in Masl